

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Medium

Question

$(4x^3 - 5x^2 + 3) - (6x^3 + 2x^2 - x)$

Which of the following expressions is equivalent to the expression above?

Answer

- A. $-10x^3 - 3x^2 + x + 3$
- B. $-2x^3 - 7x^2 + x + 3$
- C. $-2x^3 - 3x^2 + x + 3$
- D. $10x^3 - 7x^2 - x + 3$

Correct Answer: B

Rationale

Choice B is correct. Using the distributive property, the given expression can be rewritten as $4x^3 - 5x^2 + 3 - 6x^3 - 2x^2 + x$. Combining like terms, this expression can be rewritten as $(4 - 6)x^3 + (-5 - 2)x^2 + x + 3$, which is equivalent to $-2x^3 - 7x^2 + x + 3$.

Choices A, C, and D are incorrect and may result from an error when applying the distributive property or an error when combining like terms.